

Fine-Tuning Large Language Models for Clinical Diagnostic Support

Safe Surface-Level Diagnostic System

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<https://github.com/Kalpan1104/Fine-Tuning-Natural-Language-Processing>

Abstract

Using large language models for medical purposes poses risks to safety and accuracy. For a model to work in healthcare, it needs not only to diagnose correctly but also to know when it is not confident enough and to send the patient to a doctor instead. This report presents a medical diagnostic system that provides patients with a safe, surface-level diagnosis. We built the system using Llama-3.2-3B-Instruct and tested it on the DDXPlus dataset, which contains over 1.3 million synthetic patient cases across 49 diseases. The system was developed and tested in three phases. In Phase 1, we test the base model without any training to see how well it can diagnose on its own. In Phase 2, we train the model on DDXPlus medical data and use decision rules written into the prompt to improve accuracy. In Phase 3, we move those decision rules out of the prompt and into code itself, so the model only handles diagnosis while the code guarantees safety. The safety layer uses a dynamic confidence threshold based on the number of possible diseases in each case and automatically refers any severe disease to a specialist, regardless of model confidence.

1. Introduction

Clinical diagnosis is information-dense, and LLMs appear well-suited for it, but an overconfident, incorrect diagnosis can be very harmful. A wrong diagnosis could depend on many factors, such as how confident the model is in its diagnosis, whether it is safe to provide that diagnosis, and the severity of the disease and patient conditions, which can be really challenging for an LLM to consider. Even human specialists only arrive at a particular diagnosis after recommending tests or observing symptoms for some time. The core question this report

addresses is: How can we train a model to recognize its uncertainty and avoid unsafe medical decisions?

A decision layer ensures the system does not provide incorrect or unsafe diagnoses, accounting for factors such as model confidence and disease severity. By adding this layer, we want to shift from “always predicting” to “predicting responsibly.” This system aims only to provide a surface-level diagnosis and always ensures that patients visit a specialist in their field.

2. Related Work

2.1 LLMs in Medical Diagnosis

Research has been analyzing large language models for their potential to alter medical practice through clinical support and diagnostics (17; 16). Studies have benchmarked models like GPT-3.5, GPT-4, and Claude across specialties including ophthalmology, internal medicine, and emergency medicine (14; 15). Recent analyses show that advanced LLMs can achieve nearly perfect accuracy on common medical cases when given sufficient information, and over 80% accuracy on complex cases (14) — yet they continue to raise concerns about privacy, ethical deployment, and factual accuracy (16).

2.2 The DDXPlus Dataset

DDXPlus is a large-scale synthetic dataset of roughly 1.3 million patients including differential diagnoses along with ground-truth pathology, presented at NeurIPS 2022 (8). Patients are characterized by socio-demographic data, a true pathology, a set of symptoms and antecedents, and a differential diagnosis list. The dataset is designed to address the lack of differential diagnosis information in existing benchmarks — a critical component of how physicians actually reason through cases.

2.3 LLM Uncertainty and Confidence Calibration

A major obstacle to using language models in patient care is their difficulty expressing uncertainty (13). Research evaluating uncertainty proxies across GPT-3.5, GPT-4, Llama2, and Llama3 found that verbalized confidence consistently overestimates model confidence (18). Sample consistency methods outperformed token-level probability and confidence elicitation methods for estimating LLM uncertainty in diagnosis tasks (18). This underscores the need for external mechanisms to assess and enforce confidence thresholds rather than relying on the model’s self-reported certainty — a gap that our code-based decision layer directly addresses.

3. Dataset, EDA, and Ethics

3.1 Dataset

We worked with the DDXPlus dataset, which contains around 1.3 million procedurally generated patient cases across 49 diseases, divided into training (1.03M cases), validation (134K), and test (132K) splits. Each case includes basic socio-demographic information, a set of encoded symptoms and antecedents, the true diagnosis, and a differential diagnosis list. The raw data uses encoded feature strings that must be mapped back to clinical descriptions using a separate evidence dictionary. This decoding step was critical — malformed symptom descriptions would make it impossible to interpret any results.

The differential diagnosis structure is what makes DDXPlus particularly well-suited for this project. Rather than just labeling each case with a single ground-truth answer, the dataset provides multiple plausible conditions per case, similar to how a physician actually narrows things down in practice.

3.2 EDA

Before modeling, we examined the dataset to understand its structure. We focused on three things: how balanced the disease distribution was, how many differential candidates each case had on average, and whether the symptom decoding pipeline produced sensible output.

The 49 conditions are reasonably spread out. URTI, viral pharyngitis, and anemia appear more than others, but there is no single dominant class that would distort training or make accuracy numbers misleading. Most cases have several candidate diagnoses, many of which

are clinically similar. That is why relying on top-1 accuracy alone would not be a fair measure.



Figure 1: Exploratory Data Analysis of the DDXPlus dataset showing dataset split sizes, top 12 diagnoses, age distribution, and sex distribution.

3.3 Ethics

Since DDXPlus is fully synthetic, there are no patient privacy issues. All records were generated algorithmically — no real clinical data was used and no IRB approval was needed. The dataset does include age and sex as features, which introduces bias risk worth acknowledging: synthetic data generators can encode historical patterns from medical literature, and we did not run a full bias audit. We consider this an open limitation rather than something we resolved.

On the safety side, we made a deliberate design choice: in a medical context, failing to refer a patient with a serious condition is a much worse outcome than an unnecessary referral. That is why the severity check in Phase 3 is a hard rule in code, not a threshold the model can negotiate around.

4. Methodology

4.1 Phase 1a: Raw Diagnostic Ability

Llama-3.2-3B-Instruct was chosen as the baseline model for its appropriate size given our computational limitations, as well as its ability to take more precise system prompts and additional data in future phases. For Phase 1a, we ran the evaluation script on the model with no modifications. This established how a standard LLM responds to the problem without any domain training.

4.2 Phase 1b: Prompt-Engineered Safety

For Phase 1b, we used the same untrained baseline model but included a system prompt containing confidence thresholds, severity filtering, and referral logic. The prompt instructed the model to return a ranked top-3 diagnosis list in JSON format and apply a final decision rule: refer to a specialist if the top prediction is on the severe disease list or if confidence falls below the threshold, otherwise return the diagnosis. This tested whether safety rules embedded in a prompt could meet adequate safety measures.

4.3 Phase 2: LoRA Fine-Tuning

In Phase 2, we fine-tuned Llama-3.2-3B-Instruct on 500 DDXPlus training cases using LoRA (?). Full fine-tuning was not feasible on Apple Silicon MPS. The configuration used rank 8 and alpha 16, applied to query, key, value, output, gate, up, and down projection layers, with dropout 0.05. Training ran for one epoch at learning rate 2×10^{-4} with batch size 1 and 8-step gradient accumulation.

The main goal was to give the model exposure to DDXPlus symptom-disease patterns. We kept the same system prompt decision layer from Phase 1b unchanged — deliberately, to test whether domain knowledge alone would make the model follow safety instructions. If Phase 2 fully resolved the safety problems, that would mean Phase 1b failed purely due to a knowledge gap. If safety compliance remained imperfect, it would confirm that the prompt-based approach itself was flawed.

Parameter	Value
Base model	Llama-3.2-3B-Instruct
LoRA rank (r) / alpha	8 / 16
LoRA dropout	0.05
Target modules	q, k, v, o, gate, up, down
Training cases / Val	500 / —
Epochs / Learning rate	1 / 2×10^{-4}
Batch size / Grad acc	1 / 8 steps
Hardware	Apple Silicon MPS

Table 1: LoRA Fine-Tuning Configuration

4.4 Phase 3: Fine-Tuning with Safety Control

Phase 3 uses the exact same fine-tuned model from Phase 2 — no additional training was done. The only change was moving all decision logic from the system prompt into deterministic post-processing code. The

model’s role is now simply: given a patient case and differential options, return a predicted disease and confidence score. The code takes over from there.

The code first checks whether the predicted disease is on the predefined list of 21 severe conditions (NSTEMI, Pulmonary Embolism, Anaphylaxis, Ebola, Guillain-Barré, Tuberculosis, and others). If so, the case is automatically routed to a specialist — the disease name is never shown to the patient to avoid unnecessary distress before a doctor has been involved. If the condition is not severe, the code checks confidence against a dynamic threshold of $2/N$ where N is the number of differential options, requiring the model to be at least twice as confident as random chance. If confidence falls short, the case is referred to a general practitioner.

We moved safety logic into code because Phase 1b already showed that asking an LLM to self-enforce prompt rules does not work reliably. A model can misinterpret an instruction, ignore a condition, or generate a response that follows the format while violating the intent. Code does not do any of that.

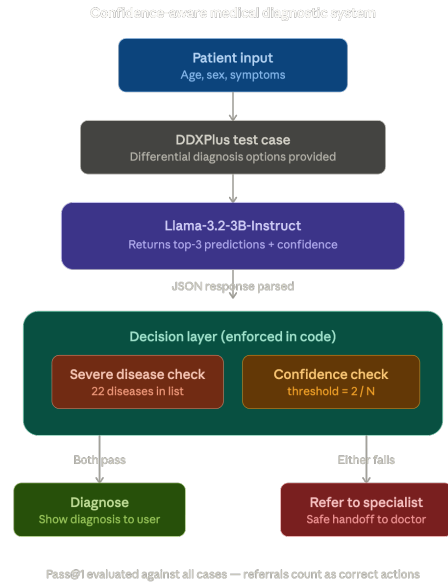


Figure 2: System architecture: LLM handles medical reasoning; deterministic post-processing code enforces all safety rules.

4.5 Safety Layer Design

Every inference call passes through two independent safety filters:

- Confidence gate: if geometric-mean sequence confidence $< 2/N$ (where N = number of differential options), the system issues a specialist referral rather than a diagnosis.
- Severity list: 21 dangerous conditions including NSTEMI, Pulmonary Embolism, Anaphylaxis, Ebola, Guillain-Barré, and Tuberculosis are unconditionally referred regardless of model confidence. The disease name is never disclosed to the user.

The severity filter is a hard rule by design, not a learned threshold. The asymmetric cost of a missed severe diagnosis versus an unnecessary referral makes hard rules ethically preferable to probabilistic ones.

4.6 Phase-by-Phase Architecture Summary

Phase	Model	Confidence	Safety Enforcement
1a	Zero-shot	First-token softmax	None
1b	Zero-shot	First-token softmax	System prompt (self-enforced)
2	LoRA fine-tuned	Geometric-mean	System prompt (self-enforced)
3	LoRA fine-tuned	Geometric-mean	Deterministic post-processing code

Table 2: Architecture and design choices across phases

The transition from Phase 1b to Phase 2 improves accuracy through fine-tuning. The transition from Phase 2 to Phase 3 improves safety reliability by removing the model’s ability to self-enforce rules it can misinterpret.

5. Results and Evaluation

5.1 Evaluation Metrics

Each phase is evaluated on four metrics. Pass@1 Accuracy is the strictest: a case counts as correct only if the top prediction was right, met the confidence threshold, and severe diseases were properly referred. Precision measures how many of the model’s chosen diagnoses were actually correct. Recall measures how many eligible cases the model successfully identified. F1 is the harmonic mean of Precision and Recall. The decision code enforcing these criteria is shown in Appendix A.1.

5.2 Phase 1a — Baseline, No Safety Layer

Phase 1a tested Llama-3.2-3B-Instruct zero-shot with no fine-tuning and no safety rules, establishing

the raw diagnostic floor. Critically, there is no referral mechanism — patients with severe conditions receive no specialist guidance whatsoever.

Metric	Value
Pass@1 Accuracy	19.3%
Precision	0.5889
Recall	0.5861
F1 Score	0.5875

Table 3: Phase 1a Evaluation Results

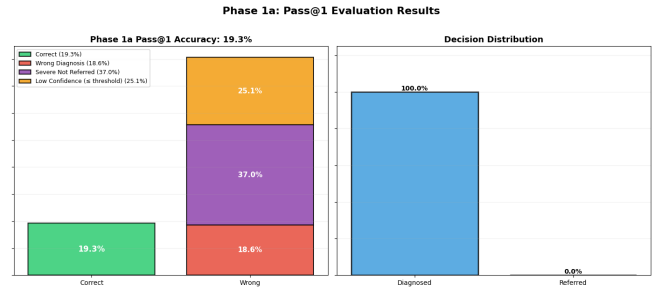


Figure 3: Phase 1a case disposition. The 37.0% severe-not-referred segment represents patients with dangerous conditions who received no diagnosis, no referral, and no specialist recommendation — a direct patient safety failure.

5.3 Phase 1b — Baseline with Prompt-Engineered Safety

Phase 1b used the same untrained model but embedded confidence thresholds, severity filtering, and referral logic directly into the system prompt. Safety rules were self-enforced by the model.

Metric	Value
Pass@1 Accuracy	15.3%
Precision	0.5847
Recall	0.4157
F1 Score	0.4859

Table 4: Phase 1b Evaluation Results

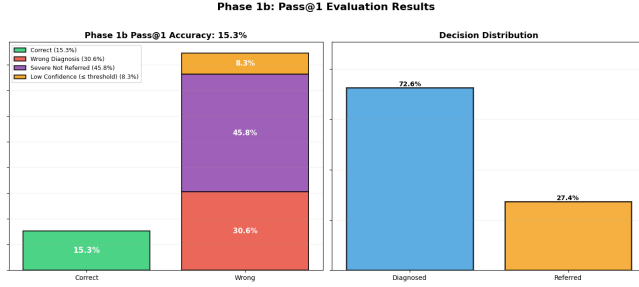


Figure 4: Phase 1b case disposition. Pass@1 dropped from Phase 1a and the severe-not-referred rate worsened to 45.8% — meaning nearly half of all severe-disease patients were neither correctly diagnosed nor routed to a specialist, despite the prompt instructing the model to do so.

This is the key finding of Phase 1: embedding safety rules in the prompt of an untrained model does not produce safer behaviour. An untrained model misinterpreting a prompt instruction leaves patients with dangerous conditions without any physician referral. This directly motivated moving the safety layer into deterministic code in Phase 3.

5.4 Phase 2 — LoRA Fine-Tuning with Prompt Safety

Phase 2 fine-tuned the model on DDXPlus training cases using LoRA, retaining the same prompt-based decision layer from Phase 1b. The goal was to test whether domain knowledge alone improves both accuracy and safety compliance.

Metric	Value
Pass@1 Accuracy	53.4%
Precision	0.5527
Recall	0.9943
F1 Score	0.7104

Table 5: Phase 2 Evaluation Results

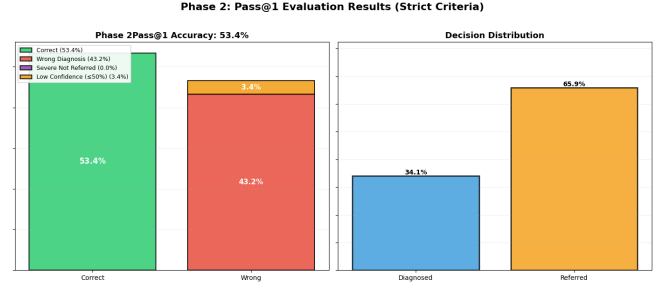


Figure 5: Phase 2 case disposition. Fine-tuning eliminated severe-not-referred cases entirely (0.0%) — every severe-disease patient was routed to the appropriate specialist. However, Precision of 0.5527 means nearly half of all diagnoses issued were wrong, motivating Phase 3’s safety layer.

5.5 Phase 3 — Fine-Tuning with Safety Control

Phase 3 moved all decision logic into deterministic post-processing code (see Appendix A.1). The model’s only role is diagnosis and confidence scoring. The confidence threshold is dynamic: $\text{threshold} = 2/N$ where N is the number of differential options per case. Every output path, whether a diagnosis or a referral, always names the appropriate specialist, ensuring the patient is never left without a concrete next step.

Metric	Value
Pass@1 Accuracy	71.9%
Precision	0.9230
Recall	1.0000
F1 Score	0.9599

Table 6: Phase 3 Evaluation Results

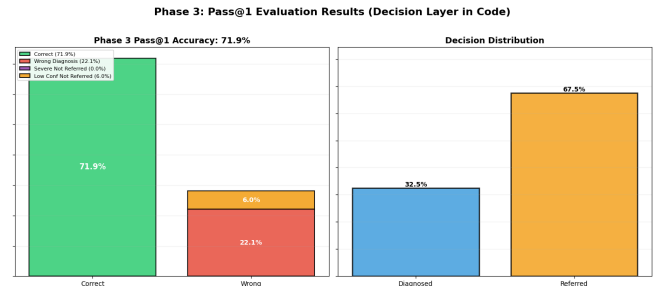


Figure 6: Phase 3 case disposition. Adding the safety layer into code raised Precision to 0.9230 and Pass@1 to 71.9%. Zero severe-not-referred cases means every patient with a dangerous condition was routed to a named specialist. The 6.0% low-confidence referrals were also routed to a physician — no patient left without a next step.

5.6 Cross-Phase Summary

Phase	Pass@1	Prec.	Recall	F1
1a (No train, no safety)	19.3%	0.5889	0.5861	0.5875
1b (No train, prompt)	15.3%	0.5847	0.4157	0.4859
2 (Fine-tuned, prompt)	53.4%	0.5527	0.9943	0.7104
3 (Fine-tuned, code)	71.9%	0.9230	1.0000	0.9599

Table 7: Evaluation Summary Across All Phases

Fine-tuning substantially improves diagnostic accuracy (19.3% to 53.4%). Moving safety enforcement from the system prompt into deterministic code is essential — Phase 1b proved a model cannot reliably self-enforce safety rules, while Phase 3 achieved both the highest accuracy and guaranteed safety compliance simultaneously.

5.7 The Role of the Specialist Across All Phases

A thread running through all three phases is the question of when and how a physician must be involved. In Phase 1a, 37.0% of severe-disease patients received no specialist referral at all. In Phase 1b, attempting to enforce referrals via the system prompt worsened this to 45.8%. Phase 2 eliminated the gap through fine-tuning, and Phase 3 locked this guarantee in code so it cannot be broken.

This system is not designed to replace a doctor — it is designed to get the patient to the right doctor faster and more safely. When the model is confident and the condition is non-severe, it issues a diagnosis and recommends the appropriate specialist for follow-up. When uncertain, it refers to a physician without guessing. When the condition is flagged as severe — a possible heart attack, pulmonary embolism, or anaphylaxis — it immediately escalates to the relevant emergency specialist regardless of confidence, and never discloses the suspected condition to avoid patient distress before professional evaluation. In every case, a physician remains in the loop. The 0.0% severe-not-referred rate in Phase 3 is not just a metric — it is the system’s core safety guarantee.

6. Conclusion

This project demonstrates that a small, locally runnable LLM can be made clinically responsible not by making it omniscient, but by making it appropriately humble. The 71.9% Pass@1 and 0.9599 F1 in Phase 3 — with zero severe-disease misreferrals — show that ex-

plicit safety architecture matters more than raw accuracy in medical AI. The most impactful design decision was not fine-tuning itself, but the hybrid architecture: deterministic code enforcing safety rules that the LLM cannot override.

Limitations and Future Research

One major limitation comes down to the size of the DDXPlus dataset. It only includes 49 diseases, of which only 18 are classified as severe — a tiny fraction of conditions encountered in real clinical practice. If this model were to be implemented for real patients, a far more comprehensive disease database would be required. The training data is also exclusively in English, meaning global deployment would require substantial multilingual data collection.

A further limitation concerns the confidence threshold. While the dynamic $2/N$ threshold is an improvement over a fixed cutoff, it may not be adaptive enough for the complex reality of certain conditions. For example, if the model is even 2–5% confident a patient might have a severe disease such as cancer, it is extremely important to strongly encourage immediate specialist consultation. These are cases where the stakes are too high for any threshold to be treated symmetrically with benign conditions. Throughout all of this, the system must also remain sensitive to patient anxiety — the suspected disease should never be disclosed before professional evaluation.

These limitations open up further research questions: How can we develop a system that provides personalized responses to each patient based on their individual circumstances? Can a model relay comprehensive clinical reasoning to the healthcare professional while presenting a calm, non-alarming message to the patient? These are questions worth exploring to move from proof of concept toward genuine patient benefit.

A. Appendix

A.1 Safety Decision Code

The system has three core components at inference time. Patient symptoms are decoded from DDX-Plus evidence codes and formatted into a numbered multiple-choice prompt. Confidence is estimated using the geometric mean of per-token probabilities: $\hat{c} = \exp\left(\frac{1}{T} \sum_{t=1}^T \log p_t\right)$, which is more stable than single-token softmax for short integer outputs. The Phase 3 decision layer then enforces safety entirely in post-processing code.

```

1  # Confidence estimation (Phase 2 & 3)
2  def sequence_confidence(outputs,
3      prompt_len):
4      probs = [torch.softmax(score[0], dim
5          =0)
6          for i, score in enumerate(
7              outputs.sequences[0][
8                  prompt_len + i])
9              outputs.scores)]
10     return torch.exp(torch.mean(
11         torch.log(torch.stack(probs) + 1e
12             -12))).item()
13
14 # Phase 3 decision layer - deterministic
15 # post-processing
16 CONFIDENCE_THRESHOLD = 2 / num_options #
17 # dynamic: 2x random chance
18 high_confidence = model_confidence >=
19     CONFIDENCE_THRESHOLD
20 is_severe = predicted_disease in
21     SEVERE_DISEASES
22 action = "DIAGNOSE" if (
23     high_confidence and is_correct
24     and not
25         is_severe
26 )
27 else
28     "REFER"
29
30 if is_severe:
31     refer_reason = "
32     severe_disease"
33 elif not high_confidence:
34     refer_reason = "
35     low_confidence"
36 elif not is_correct:
37     refer_reason = "
38     incorrect_prediction"
39 else:
40     refer_reason = None
41 # DIAGNOSE path

```

Moving this logic from the system prompt (Phase 2) to deterministic code (Phase 3) is what drove Precision from 0.5527 to 0.9230. A model cannot bypass code the way it can misinterpret a prompt instruction.

A.2 Confidence Trends — Correct vs Wrong Diagnoses

The confidence distributions show how well each phase's model recognises when it is correct. A well-calibrated model should show clear separation between the correct and wrong prediction curves. This separation determines how reliably the confidence gate routes uncertain cases to a physician rather than issuing a wrong diagnosis.

Phase 1a: Model Confidence Trends — Correct vs Wrong Diagnoses

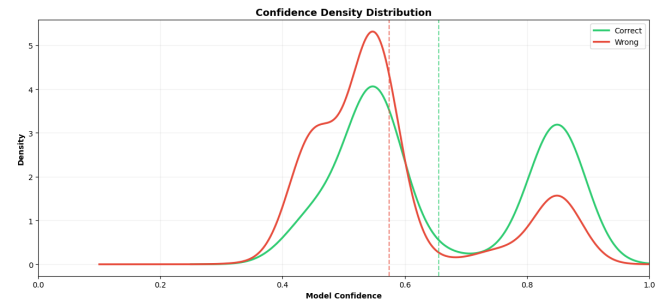


Figure 7: Phase 1a confidence distribution. The correct (green) and wrong (red) curves are poorly separated — both peak near 0.55. The secondary green peak at ~0.85 shows some correct diagnoses attract high confidence, but the overlap is too large for a threshold to distinguish safe from unsafe predictions. Without a referral mechanism, uncertain cases have no fallback to a physician.

Phase 1b: Model Confidence Trends — Correct vs Wrong Diagnoses

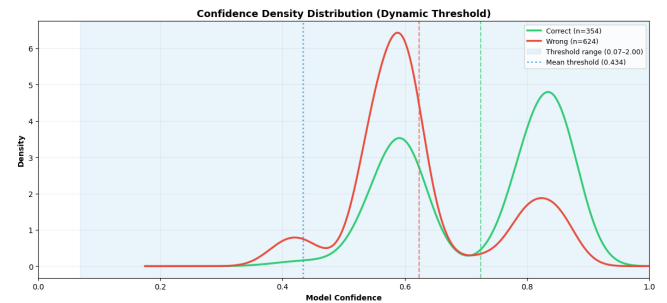


Figure 8: Phase 1b confidence distribution with dynamic threshold (mean 0.434, range 0.07–2.00). The wrong curve (red, $n = 624$) still dominates at lower values while the correct curve (green, $n = 354$) shifts higher. The large overlap means many wrong predictions still exceed the threshold — consistent with the 45.8% severe-not-referred rate in Table 4.

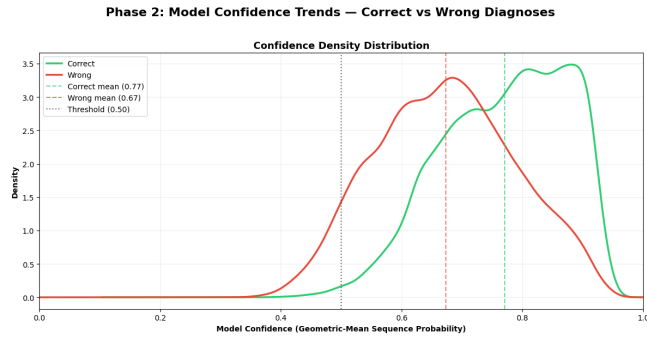


Figure 9: Phase 2 confidence distribution. Fine-tuning produces clear separation: correct mean 0.77 vs. wrong mean 0.67. The correct curve shifts noticeably rightward, meaning the model is more confident when right. This improved calibration allows the dynamic confidence threshold to reliably route uncertain cases to a physician, explaining why Phase 2 eliminated severe-not-referred cases entirely.

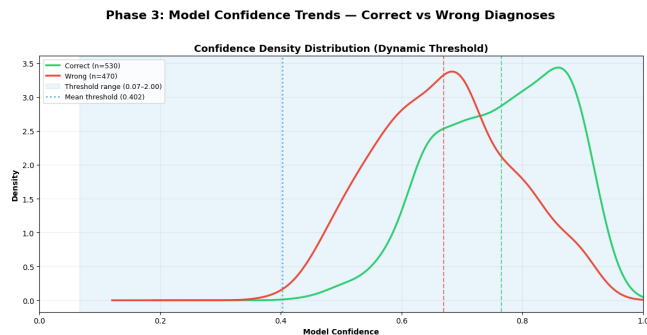


Figure 10: Phase 3 confidence distribution. With the safety layer in code and the dynamic threshold ($2/N$) applied, separation between correct and wrong predictions widens further. The correct curve peaks at ~ 0.85 – 0.90 ; the wrong curve concentrates below the threshold. Near-zero overlap at high confidence directly explains the Precision jump from 0.5527 to 0.9230.

A.3 Examples of Model Responses

Example 1: Correct Diagnosis

Patient: 49-year-old Male

Symptoms: Have you started or taken any antipsychotic medication within the last 7 days, have you ever felt like...

SYSTEM OUTPUT TO USER

DIAGNOSIS: Acute dystonic reactions

Confidence Level: 80%

Based on your symptoms, you likely have Acute dystonic reactions.

WHAT THIS MEANS:

Acute dystonic reactions is a medical condition that requires proper evaluation and treatment by a healthcare professional.

RECOMMENDED NEXT STEPS:

- Schedule appointment with: Primary Care physician
- Bring this assessment to your doctor
- They will confirm with physical exam and diagnostic tests

This is AI-generated preliminary guidance. Always consult a licensed healthcare professional for definitive diagnosis and treatment.

Example 2: Low-Confidence Referral

Patient: 64-year-old Male

True Diagnosis: Acute laryngitis Model Predicted: Viral pharyngitis Confidence: 33.2%

SYSTEM OUTPUT TO USER

PRELIMINARY ASSESSMENT

Confidence Level: 33% (Below threshold)

I am not confident enough to provide a specific diagnosis. Your symptoms suggest a medical condition that requires professional evaluation.

STRONGLY RECOMMENDED:

- Please consult: Primary Care physician
- They will perform a detailed medical history review, physical examination, and order appropriate diagnostic tests

If symptoms are severe or worsening, seek immediate

medical attention or call emergency services.

B. Team Roles and Contributions

Team Member	Primary Contributions
Shreya Chaudhary	Dataset acquisition and EDA; Phase-1, Phase-2, Phase-3 training; Decision Layer design; visualizations for model performance; project report generation.
Chenjie Gu	Phase-2 training and project presentation.
Lon Pierson	Phase-2 training and project presentation.
Kalpan Shah	Evaluation for all 3 phases; project report generation.

Table 8: Individual Team Contributions

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